

EFFICIENT AND AFFORDABLE SCHOTTKY DIODES

SiC diodes in the 600V and above range enable virtually loss-less switching in combination with good conduction behaviour, allowing high efficiency and reduced complexity in today's SMPS solutions

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Silicon carbide (SiC) as the ideal semiconductor material for power electronics applications is well known for decades. However, it was only in 2001 that first commercial devices based on SiC were introduced into the market. Reason for this long pre-development time was the difficult substrate wafer manufacturing process. In fact, the first commercial SiC devices were manufactured on a 5.1cm (2-inch) diameter wafer. Seven years later, the wafer diameter used in production was increased to 10.2 cm (4 inches). This made the formerly exotic and expensive SiC technology much more affordable.

The unique benefit of SiC diodes in the 600V and above range is that virtually loss-less switching is enabled in combination with attractive conduction behaviour, allowing benchmark efficiency and reduced complexity in today's SMPS solutions.

The third generation of SiC Schottky diodes features the industry's lowest device capacitance for any given current rating, which further enhances overall system efficiency, especially at higher switching frequencies and under low load conditions.

Optimal utilisation of SiC diodes in PFC

Which diode to choose for a given power application depends on the following factors:

1. Efficiency, both at full load and light load



Infineon's third-generation SiC Schottky diodes

2. Surge current capability, start-up and cycle drop-out
3. Thermal behaviour

Whereas the efficiency requirement clearly points towards the use of SiC Schottky barrier diodes, the surge current capability points towards a merged p-n/Schottky concept to cope with potential thermal runaway in over-current conditions. Thermal behaviour, which benefits from the improved R_{th}/Z_{th} values, has been achieved by the third generation. The only remaining task is then to choose the current rating for the given power application.

Similar to the $R_{DS(on)}$ optimisation choice for a power MOSFET, in a power-factor correction (PFC) stage the losses attributed to the diode can be split into conduction losses, which

come from the ohmic slope of the forward characteristic, and capacitive losses, which come from the capacitive displacement charge. As conduction losses decrease with a lower ohmic slope or higher current rating of the diode, capacitive losses increase with a higher current rating. As conduction losses are directly proportional to the square of the power drawn from the PFC stage, a higher current rating will be beneficial for full-load efficiency. Vice versa, a low current rating will be beneficial for light-load efficiency. Choosing the right diode is thus in the end an evaluation of how to optimise the trade-off between full-load efficiency and light-load efficiency.

Let's look at an evaluation example. Assume a 1000W continuous-conduction-mode PFC application with